

Computational Biofluid Mechanics

Patient-Specific Computational Fluid Dynamics of the Airways

Segmentation, Mesh Generation, and Flow Analysis in Tracheomalacia

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Reproducible project report; analyses still in progress are identified explicitly in the text.

Abstract

Tracheomalacia is characterised by excessive collapse or narrowing of the tracheal lumen and can substantially alter respiratory airflow. This project uses patient-specific computed tomography, three-dimensional airway segmentation, and computational fluid dynamics (CFD) to compare airflow before and after intervention. Airway surfaces are segmented in 3D Slicer, clipped along VMTK-derived centerlines, extended in the local centerline direction, and closed with planar caps. Tetrahedral volume meshes are generated with Gmsh and steady incompressible airflow is solved using OpenFOAM. This report documents the reproducible processing pipeline, evaluates mesh quality and mesh independence, and compares clinically relevant flow quantities between the preoperative and postoperative geometries. The preoperative minimum lumen area was 1.813 mm^2 , compared with 5.169 mm^2 at the matched postoperative location, giving an area constriction of 64.9%. In the steady postoperative model the local pressure drop was 30.95 Pa, the resistance was 15.41 Pa/(L min), and 73.2% of outlet flow entered the right lung. A four-level HXT mesh study selected the 0.15 mm mesh because its reported metrics differed by no more than 2.02% from the denser mesh while using 48% fewer cells. A complete-cycle transient calculation is configured; its final phase-resolved results are reported only after solver completion.

Contents

1	Introduction	1
1.1	Research questions	1
2	Critical review of a selected CFD paper	1
2.1	Goal	1
2.2	Methods	1
2.3	Results	2
2.4	Limitations	2
2.5	Evaluation	2
3	Imaging data and anatomical cases	2
4	Airway segmentation and surface preparation	3
4.1	Segmentation	3
4.2	Centerline extraction and clipping	3
4.3	Anatomical measurements and degree of constriction	3
4.4	Qualitative flow behaviour through the constriction	6
5	Volume mesh generation	6
5.1	Near-wall resolution	7
6	Mathematical model	7
6.1	Governing equations	7
6.2	Boundary conditions	7
7	Numerical method	9
7.1	Steady solver	9
7.2	Transient solver and breathing waveform	9
8	Verification	10
8.1	Mesh quality	10
8.2	Convergence and mass conservation	11
8.3	Effect of additional steady iterations	11
8.4	Mesh-independence assessment	12

9	Results	12
9.1	Velocity field	12
9.2	Pressure distribution	16
9.3	Outlet and lung flow distribution	16
9.4	Local resistance before the bifurcation	16
10	Transient postoperative analysis	19
10.1	Time-step convergence	19
10.2	Velocity vectors at representative phases	19
10.3	Breathing waveform and time-varying resistance	20
11	Discussion	20
11.1	Interpretation of outlet conditions and lung-flow split	20
11.2	Mesh sensitivity and optimal mesh	20
11.3	Constant-diameter resistance during inspiration	21
11.4	Limitations	21
11.5	Handling transient non-convergence	21
12	Conclusions	21
	Reproducibility statement	22

List of Figures

1	Case-specific airway preparation in anatomical frontal view. The top row shows the live Slicer segmentations; the bottom row shows the final CFD surfaces with centerline-directed flow extensions and planar caps.	4
2	Centerline-based anatomical correspondence. Orange curves show the measured tracheal interval and magenta contours identify the preoperative minimum and corresponding postoperative lumen sections. The postoperative location is mapped using 76.53% of inlet-to-carina centerline distance.	5
3	Qualitative velocity and static-pressure evolution through a constriction. Velocity increases as area decreases; pressure falls sharply through the stenosis and recovers only partially downstream. Curves are schematic and not CFD results.	6
4	Postoperative HXT mesh refinement in matched anatomical frontal views. The complete baseline and selected meshes are shown above, with identically framed carina close-ups below. Both levels use the same geometry, physical surface groups, and HXT algorithm; only target characteristic length differs. Surface-edge density illustrates refinement, while quantitative volume-mesh sensitivity is assessed separately in Section 8.4.	8
5	Prescribed sinusoidal inlet flow for the postoperative transient simulation. Positive and negative flow denote inspiration and expiration, respectively.	10
6	Initial residual evolution for the postoperative steady simulation. Dashed and dotted lines indicate the configured velocity and pressure convergence criteria. The SIMPLE algorithm stopped at iteration 589 after all criteria were satisfied.	12
7	Postoperative mesh-sensitivity metrics as functions of actual volume cell count. All meshes use identical geometry, boundary conditions, sampling planes, and HXT tetrahedral meshing strategy.	13
8	Absolute relative differences from the densest 0.12 mm mesh, calculated as $ \phi_i - \phi_d / \phi_d \times 100\%$. The densest case reached its 2000-iteration limit without satisfying residual control and is therefore a requested comparison rather than an accepted truth solution.	14

9	Postoperative velocity-vector field at the converged steady solution (iteration 589). Arrow direction represents local flow direction; arrow colour represents velocity magnitude on a fixed 0 m/s to 10 m/s scale. The airway surface is shown semi-transparently for anatomical context.	15
10	Postoperative dimensional gauge-pressure distribution at the converged steady solution (iteration 589), shown in anatomical frontal view. OpenFOAM kinematic pressure was multiplied by an air density of 1.204 kg/m ³ ; outlet gauge pressure is zero.	16
11	Centerline-interpolated dimensional pressure across the postoperative region corresponding to the preoperative stenosis. Distance is measured from the mapped superior limit toward the mapped inferior limit. This profile shows local pressure evolution; resistance is calculated separately from area-averaged pressures on the bounding planes.	17
12	Fixed centerline-normal planes delimiting the postoperative matched tracheal region. The blue superior plane is the upstream pressure section and the orange inferior plane is the downstream section. Their origins and normals are archived in a versioned JSON definition and reused unchanged for steady, mesh-sensitivity, and transient analyses.	18
13	Residual evolution for the postoperative transient simulation. Highlight any time steps that did not satisfy the selected criterion.	19
14	Postoperative transient velocity vectors at three phases of the breathing cycle. State each time and inlet flow rate.	19
15	Breathing-cycle flow rate and local resistance. Resistance near zero flow must be handled carefully because direct division by $Q(t)$ becomes ill-conditioned.	20

List of Tables

1	Imaging and anatomical case information.	3
2	Final segmentation settings.	3
3	Anatomical measurements obtained from centerline-normal lumen sections in 3D Slicer.	5
4	Replicable mesh settings and quality metrics. All current volume elements are first-order tetrahedra. Add the exact Gmsh version and final values from each <code>checkMesh</code> log.	7
5	OpenFOAM boundary conditions.	9
6	Software and provenance information. Replace placeholders with the immutable versions used for the final simulations.	9
7	Transient postoperative simulation settings.	11
8	Postoperative HXT mesh-quality metrics reported by <code>checkMesh</code> . The minimum cell volumes are given after conversion of the geometry from millimetres to metres.	11
9	Sensitivity of the selected axial-velocity monitor to 1000 additional steady iterations.	12
10	Mesh-independence comparison.	13
11	Outlet flow rates and fractions.	17
12	Postoperative left/right lung flow distribution.	17
13	Steady postoperative local-resistance calculation.	18

1 Introduction

Tracheomalacia involves reduced structural stability of the tracheal wall, which may cause dynamic narrowing and increased respiratory resistance. While medical imaging describes airway anatomy, CFD can quantify how geometric narrowing influences velocity, pressure loss, flow distribution, and local shear-related quantities.

The objectives of this project are to:

1. construct patient-specific preoperative and postoperative airway geometries from computed tomography;
2. establish a reproducible segmentation-to-simulation workflow;
3. quantify airflow and pressure differences between the two cases; and
4. assess whether the numerical conclusions are independent of mesh resolution.

1.1 Research questions

- How does the intervention change the minimum airway cross-section?
- How does the pressure drop across the trachea change?
- Where do peak velocities and recirculation regions occur?
- How is flow divided among the distal outlets?
- Are the reported quantities sufficiently mesh independent?

2 Critical review of a selected CFD paper

The selected paper is Taherian et al.'s patient-specific CFD evaluation of excessive dynamic airway collapse before and after tracheal stenting [1]. Each of the five required subsections below is shorter than 150 words. No paper figures are reproduced, remaining within the permitted maximum of three.

2.1 Goal

The study investigated whether patient-specific computational fluid dynamics could quantify the functional consequences of excessive dynamic airway collapse and assess improvement after tracheal stenting. The authors specifically compared inspiratory and expiratory airflow before and after intervention using CT-derived airway geometries. They aimed to determine how stenting affected pressure loss, velocity, wall shear stress, and turbulence, and whether these CFD quantities revealed clinically relevant changes that conventional pulmonary function tests might miss. The underlying question was therefore not merely whether the airway lumen became larger, but whether the anatomical intervention produced a meaningful improvement in respiratory flow behaviour.

2.2 Methods

Four patient-specific airway models represented inspiration and expiration before and after stenting. CT images were segmented in Mimics and simulated in STAR-CCM+. Each rigid model included the trachea and approximately six to eight airway generations. The authors simulated 2.5 s inspiration and 2.5 s expiration with sinusoidal resting-breathing flow. Patient-specific outlet pressure functions were derived in a preliminary step from lobar volume changes,

while the inlet was atmospheric with 10% turbulence intensity. Unsteady RANS with a low-Reynolds-number $k-\omega$ model and second-order spatial and temporal schemes was used. Inlet and outlet extensions measured one and four hydraulic diameters, respectively. Polyhedral meshes with prism layers were tested at 0.9, 1.7, and 3.2 million cells; 1.7 million cells was selected from pressure, velocity, and wall-shear-stress sensitivity.

2.3 Results

Stenting produced the clearest improvement during expiration. The severe pre-stent narrowing generated a high-velocity jet, recirculation, substantial pressure loss, and elevated wall shear stress. The reported pre-/post-stent expiratory pressure difference was approximately 26 cmH₂O, while mean tracheal wall shear stress decreased from 3.98 to 0.793 Pa, an approximately 80% reduction. Velocity variations also became smaller after stenting. Inspiratory changes were comparatively minor; mean wall shear stress increased from 0.0354 to 0.0813 Pa, and the stent introduced some local flow disturbance. Turbulence intensity increased in parts of both phases after intervention. Spirometry changed only modestly—FEV₁ increased by 290 mL—whereas CFD exposed pronounced expiratory functional improvement that pulmonary function testing did not clearly reflect.

2.4 Limitations

The study considered only one patient, so its quantitative findings cannot be generalised to all patients with excessive dynamic airway collapse. The airway walls were rigid even though wall deformation is the defining feature of the disease. Separate end-inspiration and end-expiration geometries represented the extremes of breathing rather than continuously moving anatomy; fluid–structure interaction was omitted because patient-specific tissue properties were unavailable. The upper airway and deeper peripheral generations were truncated, although both may influence tracheal flow and outlet impedance. Outlet pressures were inferred indirectly from lobar volume changes, and local CFD predictions were not validated against invasive pressure or velocity measurements. The RANS turbulence model and selected boundary conditions add model-form uncertainty. Finally, mesh sensitivity was demonstrated primarily for selected quantities in one pre-stent expiratory geometry.

2.5 Evaluation

For this patient, the authors largely answered their research question. The paired pre-/post-stent and inspiration/expiration design isolated the phase in which airway collapse mattered most, and pressure, velocity, and wall shear stress changed coherently after intervention. The contrast with relatively insensitive spirometry supports the claim that CFD can provide additional functional information. The mesh test and patient-informed outlet conditions strengthen the internal comparison. However, the study establishes a patient-specific proof of concept rather than a validated diagnostic or prognostic method. Rigid endpoint geometries cannot reproduce dynamic coupling between airflow and collapse, and a single case cannot establish clinical accuracy or treatment thresholds. Thus, the conclusion that stenting improved this patient’s expiratory airflow is supported, whereas broader claims about CFD-guided management require validation in more patients with dynamic or fluid–structure-interaction models.

3 Imaging data and anatomical cases

Two computed-tomography datasets are analysed: a preoperative case and a postoperative case. Table 1 should be completed from the DICOM metadata and segmentation logs.

Table 1: Imaging and anatomical case information.

Case	Voxel dimensions	Voxel spacing (mm)	Clinical state
Preoperative	[insert]	[insert]	Before intervention
Postoperative	[insert]	[insert]	After intervention

All patient-derived data must be handled according to the applicable privacy, ethics, and data-management requirements. No directly identifying information should appear in this report or repository.

4 Airway segmentation and surface preparation

4.1 Segmentation

The CT series is loaded into 3D Slicer. A case-specific seed is placed in the airway lumen, and connected region growth is performed within a selected Hounsfield-unit interval. Thresholds and seed coordinates are stored under `segmentation/assets/<case>/`. The result is inspected in axial, coronal, sagittal, and three-dimensional views to exclude lung parenchyma and external air while retaining the clinically relevant branches.

Record the final settings in Table 2.

Table 2: Final segmentation settings.

Case	Lower HU	Upper HU	Segmented voxels
Preoperative	[insert]	[insert]	[insert]
Postoperative	[insert]	[insert]	[insert]

4.2 Centerline extraction and clipping

A centerline network is calculated using VMTK. Visually refined clipping points are loaded as the Slicer markup node `AirwayCutEndpoints`. The airway is clipped using the same *Clip Vessel* logic as the SlicerVMTK user interface, with `AirwayNetworkModel` providing the centerline direction.

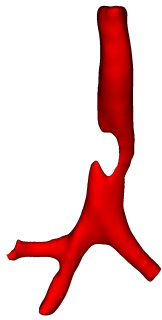
Flow extensions are aligned with the local centerline and adapt to the airway radius at each clipping location. Planar caps close the extensions for CFD. The final Slicer model is `AirwayExtendedSurfaceCapped` and is exported as `meshes/<case>/airways.stl`, preventing the two anatomies from silently overwriting one another.

4.3 Anatomical measurements and degree of constriction

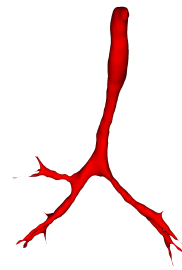
The preoperative constriction is bounded by proximal and distal landmarks on the tracheal centerline. Its length is measured along that centerline. Cross-sections normal to the local centerline are sampled through the narrowed region to locate the minimum lumen area. Because the lumen need not be circular, an equivalent circular diameter is calculated from section area A as

$$D_{\text{eq}} = 2\sqrt{\frac{A}{\pi}}. \quad (1)$$

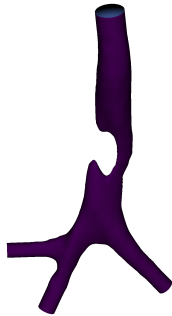
The postoperative area and equivalent diameter are measured using the same procedure at 76.53% of the inlet-to-carina centerline length. The carina is defined from the airway-network bifurcation nearest the airway seed. The primary area-based degree of constriction is



(a) Preoperative cleaned airway segmentation.



(b) Postoperative airway segmentation.



(c) Preoperative clipped, extended, and capped surface.



(d) Postoperative clipped, extended, and capped surface.

Figure 1: Case-specific airway preparation in anatomical frontal view. The top row shows the live Slicer segmentations; the bottom row shows the final CFD surfaces with centerline-directed flow extensions and planar caps.

$$C_A = \left(1 - \frac{A_{\text{pre,min}}}{A_{\text{post,matched}}}\right) \times 100\%. \quad (2)$$

The measured area constriction is 64.9%. For transparency, the corresponding minimum-Feret-diameter reduction is 65.2% and is reported as

$$C_D = \left(1 - \frac{D_{\text{pre,min}}}{D_{\text{post,matched}}}\right) \times 100\%. \quad (3)$$

Table 3: Anatomical measurements obtained from centerline-normal lumen sections in 3D Slicer.

Quantity	Symbol	Value
Preoperative constriction length	L_{pre}	12.157 mm
Preoperative minimum area	$A_{\text{pre,min}}$	1.813 mm ²
Preoperative minimum Feret diameter	$D_{\text{pre,Feret}}$	0.798 mm
Preoperative equivalent diameter	$D_{\text{pre,eq}}$	1.519 mm
Postoperative matched area	$A_{\text{post,matched}}$	5.169 mm ²
Postoperative minimum Feret diameter	$D_{\text{post,Feret}}$	2.292 mm
Postoperative equivalent diameter	$D_{\text{post,eq}}$	2.565 mm
Area constriction	C_A	64.9%
Minimum Feret diameter reduction	$C_{D,\text{Feret}}$	65.2%

The measured dimensions are consistent with the scale reported in paediatric tracheal-surgery literature. Arcieri et al. described 25 children with congenital tracheal stenosis and reported a mean internal diameter of 1.5 mm; the cohort had a median operative age of seven months and median weight of 5.2 kg [2]. This closely matches the 1.519 mm preoperative equivalent diameter obtained here. The smaller minimum Feret diameter of 0.798 mm is not contradictory: it characterises the narrowest caliper dimension of a non-circular lumen, whereas the equivalent diameter is derived from total cross-sectional area. The literature comparison supports dimensional plausibility but is not an independent validation of the patient-specific segmentation or landmark placement.

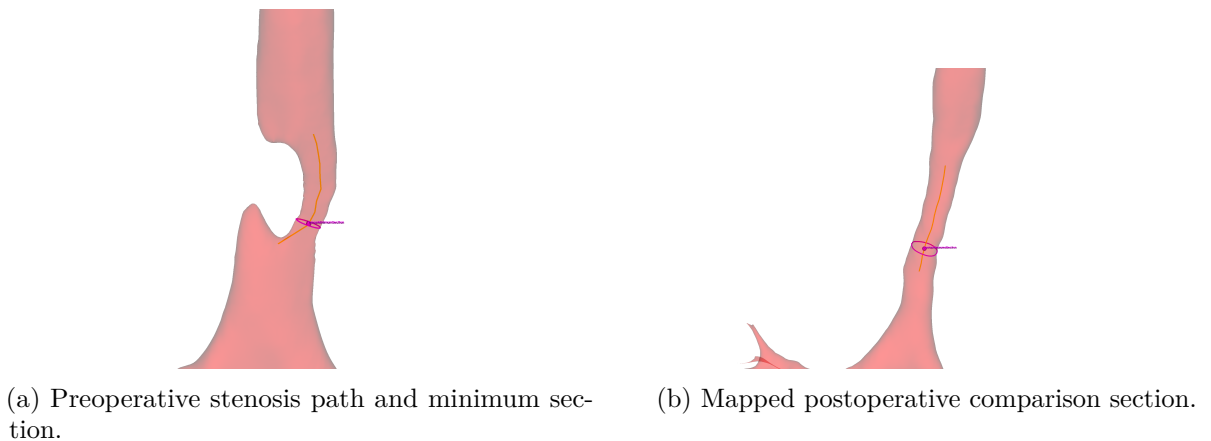


Figure 2: Centerline-based anatomical correspondence. Orange curves show the measured tracheal interval and magenta contours identify the preoperative minimum and corresponding postoperative lumen sections. The postoperative location is mapped using 76.53% of inlet-to-carina centerline distance.

4.4 Qualitative flow behaviour through the constriction

For incompressible flow, continuity $Q = AU$ requires velocity to rise as the area narrows and fall after expansion. Before the stenosis, pressure decreases gradually because viscous wall shear opposes motion; for laminar circular flow, $\Delta p = 128\mu LQ/(\pi D^4)$. Through the stenosis, acceleration converts static pressure into kinetic energy according to Bernoulli's principle, while viscosity and separation add irreversible loss. Downstream, deceleration causes partial pressure recovery, but pressure remains below the upstream trend because separation, mixing, and wall friction dissipate mechanical energy.

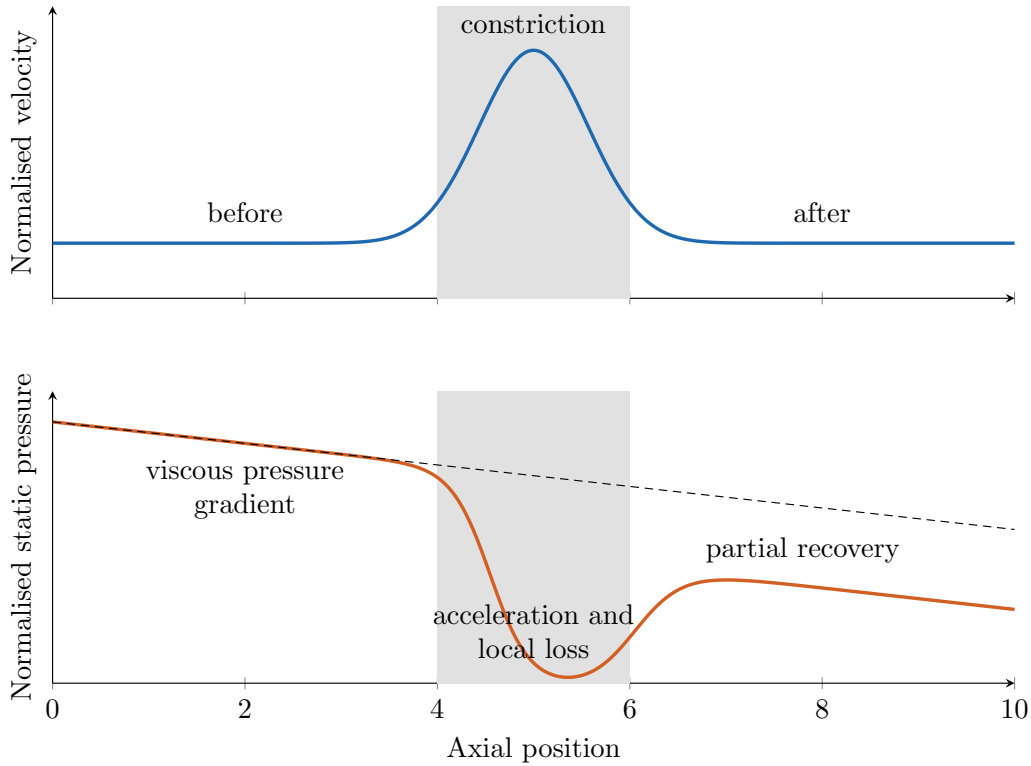


Figure 3: Qualitative velocity and static-pressure evolution through a constriction. Velocity increases as area decreases; pressure falls sharply through the stenosis and recovers only partially downstream. Curves are schematic and not CFD results.

5 Volume mesh generation

The triangulated STL surface is imported into Gmsh. Surface classification and geometry creation produce a closed fluid volume. The physical boundary groups are:

- `inlet`: tracheal inlet;
- `outlet_1`: right superior lobar bronchus;
- `outlet_2`: right inferior lobar bronchus;
- `outlet_3`: left main bronchus;
- `wall`: remaining airway surface; and
- `fluid`: internal volume.

The physical groups are visually verified whenever the STL topology or Gmsh classification changes. The volume consists of first-order tetrahedra with a uniform target characteristic length controlled by `MESH_SIZE`.

Table 4: Replicable mesh settings and quality metrics. All current volume elements are first-order tetrahedra. Add the exact Gmsh version and final values from each `checkMesh` log.

Mesh	Target size (mm)	Points	Cells	Element type	Max. aspect ratio	Max. non-orthogonality
HXT baseline	0.25	39,977	180,163	Tetrahedron	114.14	88.63°
HXT fine 1	0.20	71,198	339,886	Tetrahedron	11.73	80.29°
HXT fine 2	0.15	151,599	776,568	Tetrahedron	12.63	69.82°
HXT densest	0.12	281,330	1,493,440	Tetrahedron	88.00	82.34°

5.1 Near-wall resolution

Boundary-layer prism elements are commonly used near no-slip walls because they resolve steep wall-normal velocity gradients efficiently and with better cell alignment than isotropic tetrahedra. The current Gmsh workflow uses only first-order tetrahedra and does not include dedicated inflation layers. This must not be presented as a hybrid boundary-layer mesh. Its consequences are considered when interpreting wall-adjacent quantities, and adding prism layers is identified as a model improvement.

The Gmsh mesh uses millimetres because it inherits the Slicer coordinate scale. After `gmshToFoam`, all coordinates are multiplied by 10^{-3} to obtain SI units.

6 Mathematical model

6.1 Governing equations

Air is modelled as an incompressible Newtonian fluid. For steady flow, mass and momentum conservation are:

$$\nabla \cdot \mathbf{u} = 0, \quad (4)$$

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}, \quad (5)$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and μ is dynamic viscosity. OpenFOAM uses kinematic pressure and the kinematic viscosity $\nu = \mu/\rho$. The configured value is $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$.

The current case uses the laminar model and the steady-state `simpleFoam` solver. The appropriateness of the laminar assumption should be assessed using a Reynolds number based on representative velocity U and diameter D :

$$\text{Re} = \frac{UD}{\nu}. \quad (6)$$

[Calculate Reynolds numbers at the inlet and stenosis and discuss whether a laminar, transitional, or turbulent model is justified.]

6.2 Boundary conditions

A prescribed volumetric flow rate is applied at the tracheal inlet:

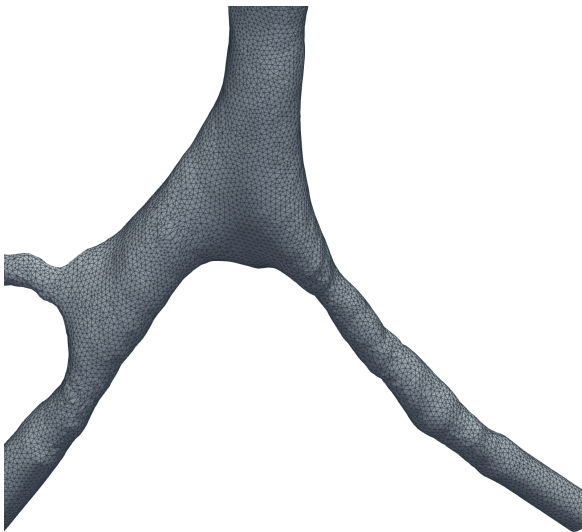
$$Q_{\text{in}} = 3.3333 \times 10^{-5} \text{ m}^3/\text{s}. \quad (7)$$



(a) 0.25 mm HXT baseline; 180,163 volume cells.



(b) Selected 0.15 mm HXT mesh; 776,568 volume cells.



(c) Baseline surface triangulation at the carina.



(d) Selected-mesh surface triangulation at the carina.

Figure 4: Postoperative HXT mesh refinement in matched anatomical frontal views. The complete baseline and selected meshes are shown above, with identically framed carina close-ups below. Both levels use the same geometry, physical surface groups, and HXT algorithm; only target characteristic length differs. Surface-edge density illustrates refinement, while quantitative volume-mesh sensitivity is assessed separately in Section 8.4.

This corresponds to approximately 2 L/min, representative of the stated minute volume for a 10-month-old infant. A no-slip condition is applied at the airway wall. The three distal outlets use the same zero gauge pressure with pressure-compatible velocity conditions. No outlet flow fraction is prescribed; consequently, the flow division is an output of the CFD solution. For a common downstream pressure, each pathway behaves approximately as

$$Q_i \sim \frac{\Delta P}{R_i}, \quad (8)$$

where differences in effective resistance R_i arise from features such as branch diameter, length, branching angle, curvature, and flow separation. Table 5 summarises the implementation.

Table 5: OpenFOAM boundary conditions.

Patch	Velocity $\mathbf{u}$	Pressure p
inlet	flowRateInletVelocity	zeroGradient
outlet_1-3	pressureInletOutletVelocity	fixedValue 0
wall	noSlip	zeroGradient

7 Numerical method

The teaching staff approved the use of 3D Slicer and SlicerVMTK for geometry preparation, Gmsh for meshing, OpenFOAM for CFD, and independent local/remote compute infrastructure. This changes the software implementation but not the required physical analyses or validation.

Table 6: Software and provenance information. Replace placeholders with the immutable versions used for the final simulations.

Stage	Software	Version or identifier
Segmentation	3D Slicer + SlicerVMTK	5.12.2; extension [insert]
Surface/volume meshing	Gmsh	[insert]
CFD	OpenFOAM Docker image	2512; image ID [insert]
Post-processing	ParaView	[insert]
Source state	Git	commit [insert]

7.1 Steady solver

The finite-volume discretisation is implemented in OpenFOAM 2512. Pressure is solved using GAMG and velocity using a symmetric Gauss–Seidel smoothed solver. The SIMPLE algorithm uses two non-orthogonal correctors. The configured residual controls are 10^{-5} for pressure and 10^{-6} for velocity.

The parallel mesh is decomposed using the Scotch method. Simulations are run in the `opencfd/openfoam-default:latest` Docker image. The exact image ID and Git commit are recorded in Table 6.

7.2 Transient solver and breathing waveform

The selected 0.15 mm HXT mesh is reused for the postoperative transient simulation. The incompressible, Newtonian equations are solved with `pimpleFoam`. A symmetric sinusoidal

volumetric-flow boundary condition is prescribed at the tracheal inlet,

$$Q(t) = Q_{\max} \sin\left(\frac{2\pi t}{T}\right), \quad (9)$$

where $T = 2.0$ s and $Q_{\max} = 1.0472 \times 10^{-4}$ m³/s (6.283 L/min). The positive half-cycle represents inspiration and the negative half-cycle expiration. Integrating either half-cycle gives a tidal volume of 66.7 mL, consistent with the assumed 2 L/min minute ventilation at 30 breaths per minute.

The inlet uses `flowRateInletVelocity`. All three outlets have zero fixed kinematic pressure and `pressureInletOutletVelocity`, which permits flow reversal during expiration. The rigid airway wall remains no-slip. Time is discretised with the second-order backward scheme. Three PIMPLE outer correctors, two pressure correctors, and one non-orthogonal corrector are used. An initial time step of 1×10^{-5} s is adjusted subject to $Co_{\max} = 2$ and a maximum step of 5×10^{-5} s; fields are written every 0.02 s. The original $Co_{\max} = 1$ timing test was rejected because a small subset of cells forced the step below 1×10^{-5} s while the mean Courant number remained below 0.08. A 0.05 s timing run and a subsequent pilot through 0.55 s precede the complete cycle. Pilot results determine whether these controls are adequate; periodicity cannot be claimed from a single cycle and will be assessed by comparing corresponding quantities in consecutive cycles if an additional cycle is required.

At the estimated peak mean velocity of approximately 20.3 m/s in the matched section, the Mach number is about 0.06, supporting the incompressibility approximation. The corresponding Reynolds number is approximately 3500, so transitional effects remain a model limitation. A shorter, more realistic mechanically ventilated inspiratory phase would require a larger peak flow for the same tidal volume and could approach the commonly used $Ma = 0.1$ compressibility threshold.

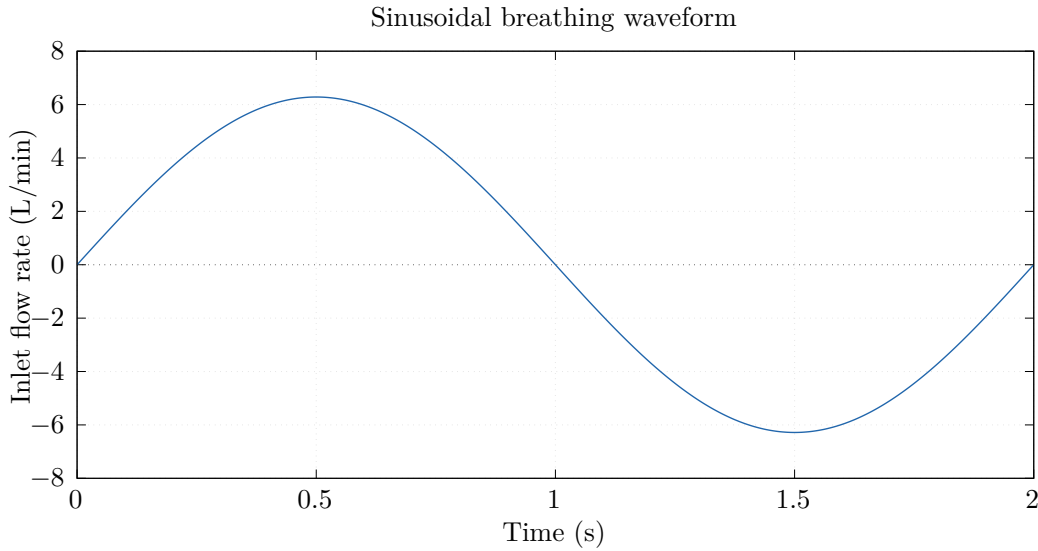


Figure 5: Prescribed sinusoidal inlet flow for the postoperative transient simulation. Positive and negative flow denote inspiration and expiration, respectively.

8 Verification

8.1 Mesh quality

Table 8 summarises the principal `checkMesh` metrics for the four postoperative HXT meshes. All meshes contained one connected fluid region. A face was counted as severely non-orthogonal

Table 7: Transient postoperative simulation settings.

Parameter	Value
Solver	<code>pimpleFoam</code>
Breathing-cycle duration	2.0 s
Initial / maximum time step	1×10^{-5} s / 5×10^{-5} s
Maximum Courant number	2.0
Field write interval	0.02 s
Timing / pilot intervals	0.05 s / 0.55 s
Simulated cycles	One initially; extend if not periodic
PIMPLE correctors	Up to 3 outer, 2 pressure, 1 non-orthogonal
Outer stopping targets	2×10^{-3} (U), 10^{-2} (p)
Linear-equation tolerance	10^{-8} (U and p)
Parallel decomposition	Scotch, 48 MPI ranks

when its non-orthogonality exceeded 70° .

Table 8: Postoperative HXT mesh-quality metrics reported by `checkMesh`. The minimum cell volumes are given after conversion of the geometry from millimetres to metres.

Target size	Max. aspect ratio	Max. non-orthogonality	Severe faces	Max. skewness	Min. volume (m ³)	Regions
0.25 mm	114.14	88.63°	2	1.457	7.90×10^{-15}	1
0.20 mm	11.73	80.29°	2	1.268	3.64×10^{-14}	1
0.15 mm	12.63	69.82°	0	1.196	5.12×10^{-14}	1
0.12 mm	88.00	82.34°	19	1.396	8.71×10^{-16}	1

The 0.15 mm mesh had the strongest overall quality: its maximum non-orthogonality remained below 70° , it contained no severely non-orthogonal faces, and it had the lowest maximum skewness of the series. Its maximum aspect ratio of 12.63 was also substantially lower than those of the 0.25 mm and 0.12 mm meshes. Refining the target size did not improve every quality measure monotonically. In particular, the 0.12 mm mesh introduced 19 severely non-orthogonal faces, a maximum aspect ratio of 88.00, and the smallest cell volume in the series. This local deterioration is consistent with its poorer solver convergence and shows why cell count alone is not a sufficient mesh-selection criterion. Nevertheless, all four meshes remained single connected domains and their maximum skewness was below the OpenFOAM internal-cell warning limit of 4. The quality results therefore support selection of the 0.15 mm mesh for the final analysis.

8.2 Convergence and mass conservation

The SIMPLE solution met residual control after 589 iterations. Final initial residuals were 9.93×10^{-7} , 7.57×10^{-7} , and 6.48×10^{-7} for U_x , U_y , and U_z , respectively, and 4.92×10^{-6} for pressure. These are below the configured criteria of 10^{-6} for velocity and 10^{-5} for pressure. Final outlet-flow integration is still used to verify that

$$Q_{\text{in}} + \sum_{i=1}^3 Q_{\text{out},i} \approx 0, \quad (10)$$

using a consistent outward-normal sign convention. Report the relative mass imbalance.

8.3 Effect of additional steady iterations

Assignment 4 requests an assessment of whether 1000 additional iterations alter an axial-velocity parameter. A fixed centerline-normal section before the bifurcation is used, and its area-averaged

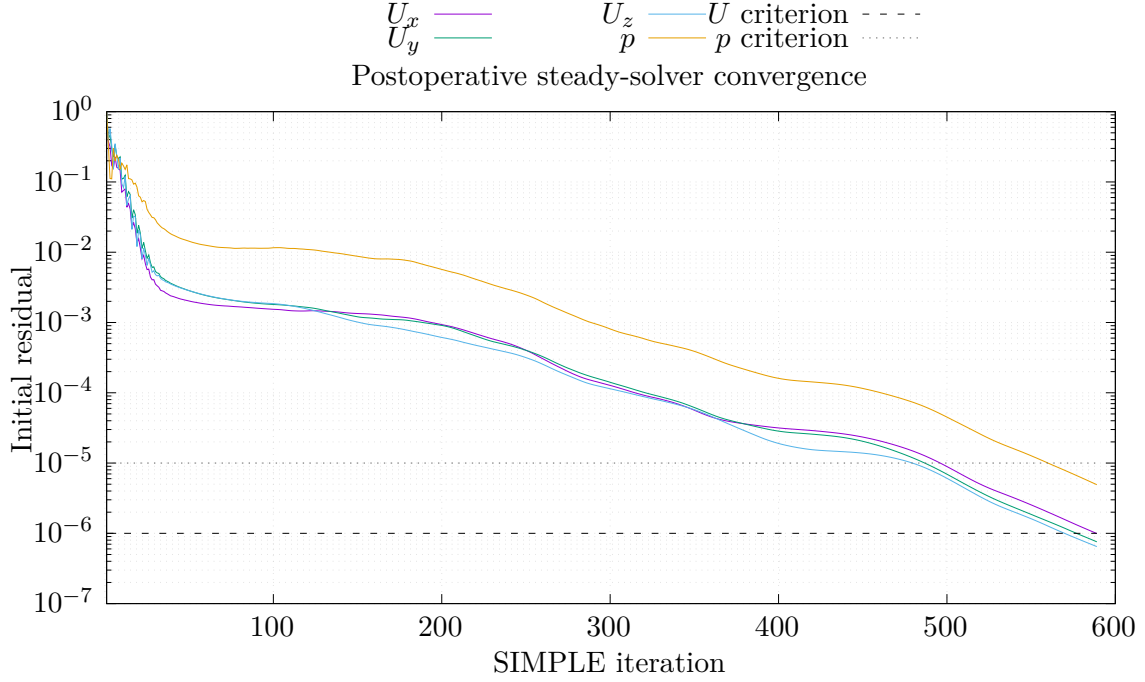


Figure 6: Initial residual evolution for the postoperative steady simulation. Dashed and dotted lines indicate the configured velocity and pressure convergence criteria. The SIMPLE algorithm stopped at iteration 589 after all criteria were satisfied.

axial velocity is recorded before and after continuation without reinitialising the solution.

Table 9: Sensitivity of the selected axial-velocity monitor to 1000 additional steady iterations.

Parameter	Iteration 2000	Iteration 3000	Relative change (%)
Area-averaged axial velocity	[insert]	[insert]	[insert]

8.4 Mesh-independence assessment

At least three postoperative mesh levels are preferred so that a trend can be assessed rather than only a pairwise difference. Each parameter of interest is plotted against the number of volume elements. For mesh i and the densest mesh d , the Assignment 5 percentage difference is defined as

$$\delta_{\phi,i} = \frac{|\phi_i - \phi_d|}{|\phi_d|} \times 100\%. \quad (11)$$

9 Results

9.1 Velocity field

At the converged postoperative solution, velocity vectors accelerate through the narrow distal trachea and divide at the carina. The fixed matched section has an area-averaged axial velocity of 6.53 m/s and a sectional peak magnitude of 9.25 m/s. The vectors also show the unequal right/left flow division quantified from patch fluxes.

Table 10: Mesh-independence comparison.

Quantity	0.25 mm	0.20 mm	0.15 mm	0.12 mm
Resistance (Pa/(L/min))	15.153	16.361	17.544	17.906
Peak velocity (m/s)	9.447	9.573	9.670	9.822
Right-lung fraction (%)	73.664	73.619	73.686	73.397
Right-superior share (%)	15.231	15.597	16.017	16.250
Solver iterations	596	770	1611	2000*

*Iteration limit reached without satisfying residual control.

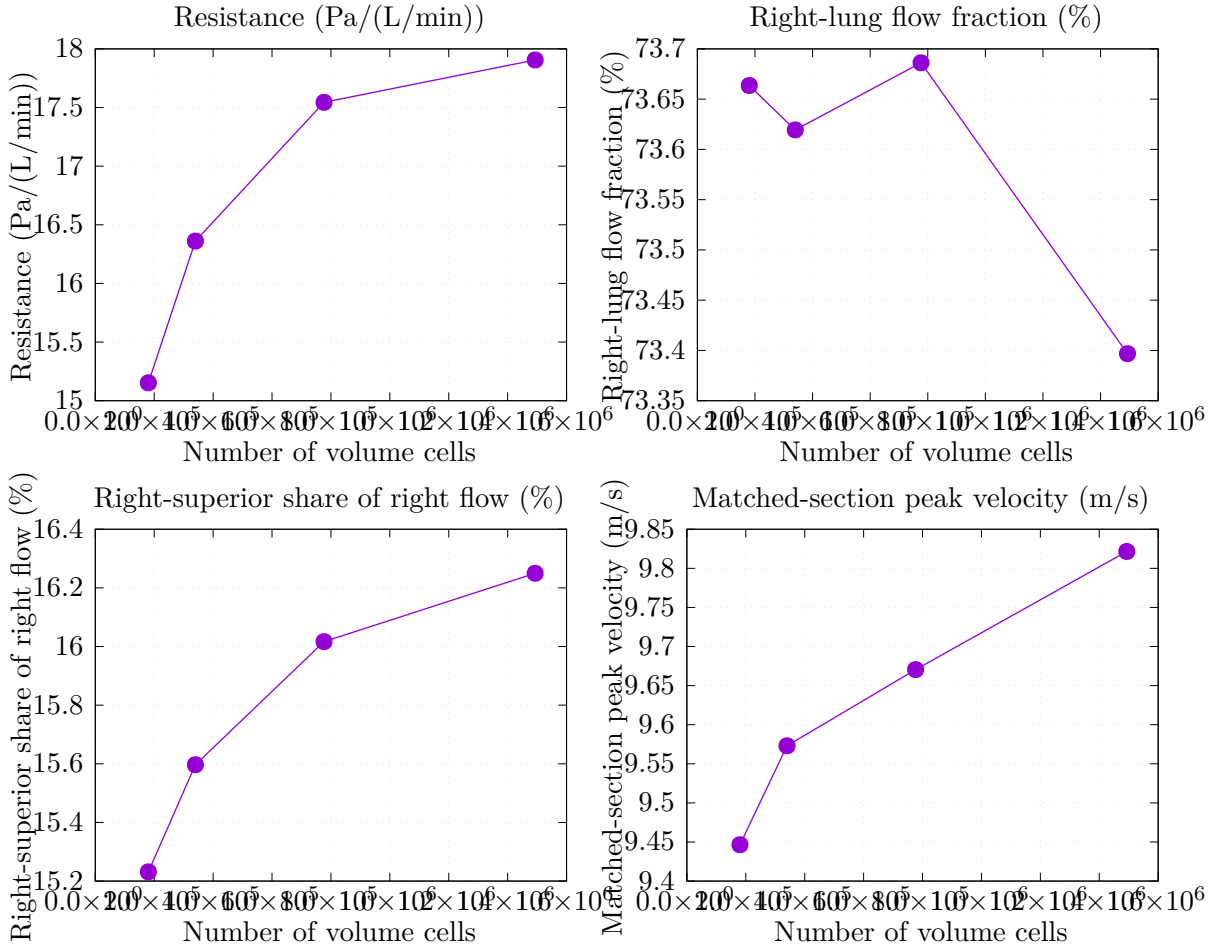


Figure 7: Postoperative mesh-sensitivity metrics as functions of actual volume cell count. All meshes use identical geometry, boundary conditions, sampling planes, and HXT tetrahedral meshing strategy.

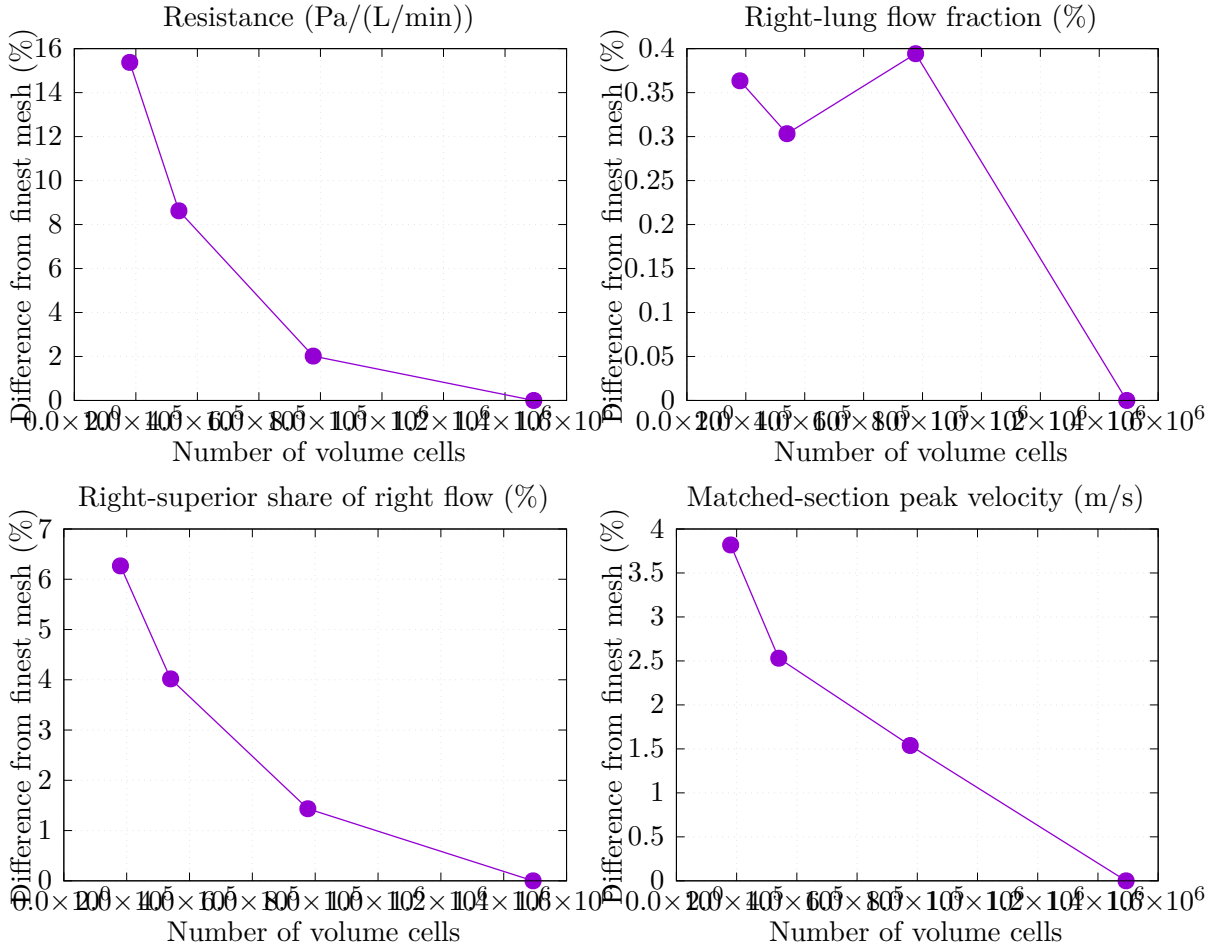


Figure 8: Absolute relative differences from the densest 0.12 mm mesh, calculated as $|\phi_i - \phi_d|/|\phi_d| \times 100\%$. The densest case reached its 2000-iteration limit without satisfying residual control and is therefore a requested comparison rather than an accepted truth solution.



Figure 9: Postoperative velocity-vector field at the converged steady solution (iteration 589). Arrow direction represents local flow direction; arrow colour represents velocity magnitude on a fixed 0 m/s to 10 m/s scale. The airway surface is shown semi-transparently for anatomical context.

9.2 Pressure distribution

Pressure is reported relative to the zero-gauge-pressure outlets and converted from OpenFOAM kinematic pressure using $P = \rho p$ with $\rho = 1.204 \text{ kg/m}^3$. Pressure decreases from the tracheal inlet toward the outlets. Along the mapped centerline interval it falls from approximately 65.1 Pa to 34.1 Pa, with a local minimum followed by partial recovery. The resistance calculation uses area-averaged plane pressures rather than centerline point samples.

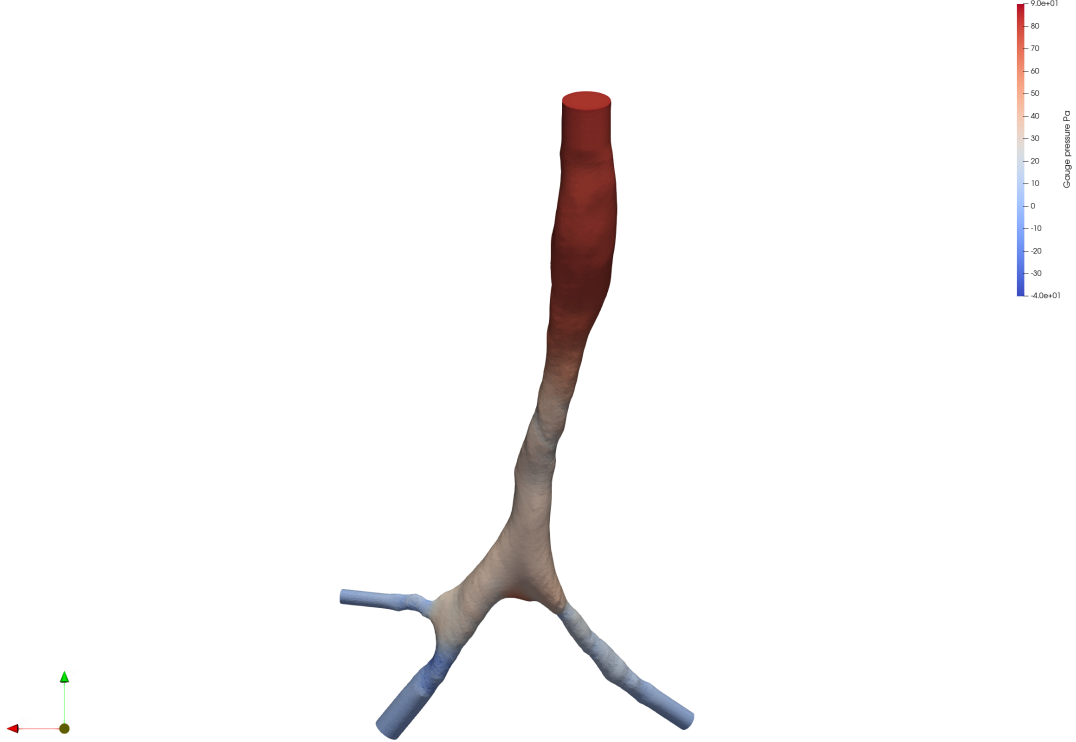


Figure 10: Postoperative dimensional gauge-pressure distribution at the converged steady solution (iteration 589), shown in anatomical frontal view. OpenFOAM kinematic pressure was multiplied by an air density of 1.204 kg/m^3 ; outlet gauge pressure is zero.

9.3 Outlet and lung flow distribution

Outlet flow was obtained by integrating outward-normal volumetric flux over each patch. The right superior and inferior lobar outlets were grouped as the right lung; the left main bronchus formed the left group. The primary left/right result was

$$Q_R = 73.17\%, \quad Q_L = 26.83\%. \quad (12)$$

The individual outlet values in Table 11 provide the supporting lobar decomposition. Because identical zero gauge pressure was imposed at all outlets, no 73/27 split was prescribed. The unequal distribution instead reflects the geometric and hydraulic resistance of the branches resolved within the truncated airway domain.

9.4 Local resistance before the bifurcation

Two planes normal to the centerline delimit a fixed section before the bifurcation, corresponding approximately to the location of the preoperative constriction. These same plane definitions are

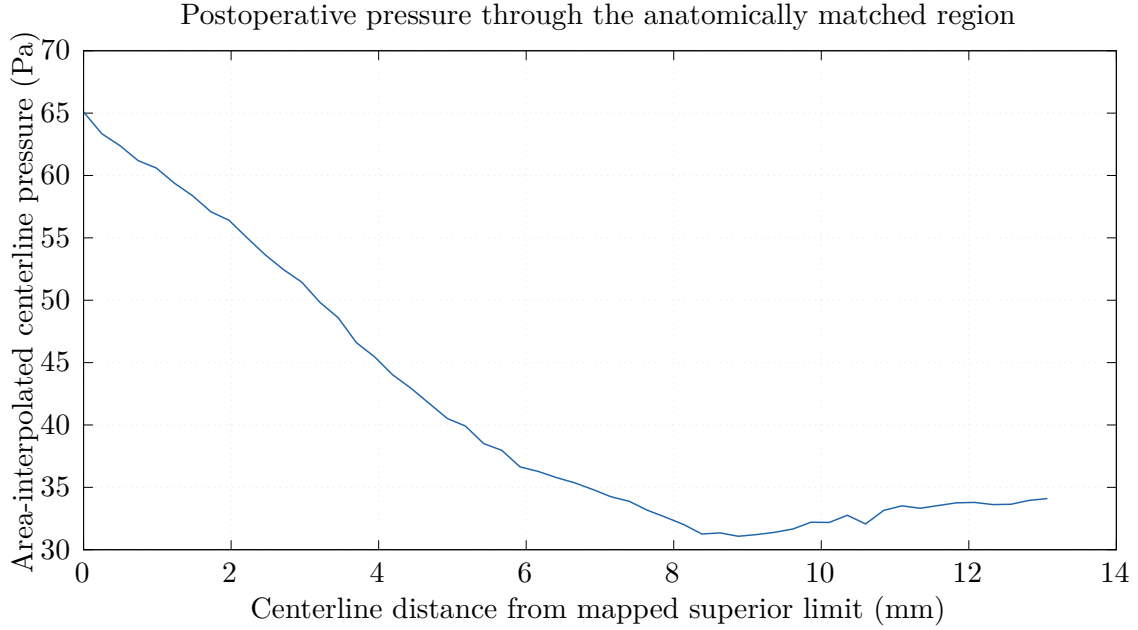


Figure 11: Centerline-interpolated dimensional pressure across the postoperative region corresponding to the preoperative stenosis. Distance is measured from the mapped superior limit toward the mapped inferior limit. This profile shows local pressure evolution; resistance is calculated separately from area-averaged pressures on the bounding planes.

Table 11: Outlet flow rates and fractions.

Case	Outlet 1 (%)	Outlet 2 (%)	Outlet 3 (%)	Imbalance (%)
Postoperative	11.23	61.94	26.83	6.0×10^{-7}

Table 12: Postoperative left/right lung flow distribution.

Lung	Flow rate (m^3/s)	Fraction (%)
Left	8.9449×10^{-6}	26.83
Right	2.4388×10^{-5}	73.17

reused for every mesh and for the transient Assignment 6 analysis. With dimensional pressure difference ΔP and volumetric flow Q , local resistance is

$$R(t) = \frac{\overline{P}_{\text{up}}(t) - \overline{P}_{\text{down}}(t)}{Q(t)}. \quad (13)$$

For the incompressible OpenFOAM pressure field p , $\Delta P = \rho \Delta p$. Area-averaged kinematic pressures were $53.70 \text{ m}^2/\text{s}^2$ upstream and $27.99 \text{ m}^2/\text{s}^2$ downstream. With $\rho = 1.204 \text{ kg}/\text{m}^3$, the local pressure drop was 30.95 Pa ; at $Q = 3.347 \times 10^{-5} \text{ m}^3/\text{s}$ this gives $R = 9.25 \times 10^5 \text{ Pa s}/\text{m}^3$, equivalent to $15.41 \text{ Pa}/(\text{L min})$. The matched-plane mean axial and peak velocities were 6.53 m/s and 9.25 m/s , respectively.



Figure 12: Fixed centerline-normal planes delimiting the postoperative matched tracheal region. The blue superior plane is the upstream pressure section and the orange inferior plane is the downstream section. Their origins and normals are archived in a versioned JSON definition and reused unchanged for steady, mesh-sensitivity, and transient analyses.

Table 13: Steady postoperative local-resistance calculation.

Quantity	Value
Upstream area-averaged kinematic pressure	$53.695 \text{ m}^2/\text{s}^2$
Downstream area-averaged kinematic pressure	$27.993 \text{ m}^2/\text{s}^2$
Pressure difference ΔP	30.945 Pa
Volumetric flow Q	$3.347 \times 10^{-5} \text{ m}^3/\text{s}$
Resistance $R = \Delta P/Q$	$9.247 \times 10^5 \text{ Pa s}/\text{m}^3$

10 Transient postoperative analysis

10.1 Time-step convergence

At the time of report assembly, the full 2 s calculation on the selected 0.15 mm HXT mesh was running on 48 MPI ranks. Startup tests showed that adaptive stepping maintained the maximum Courant number close to the target of 2 while the mean remained substantially lower. Final linear-solver residuals were typically of order 10^{-9} and final global continuity errors of order 10^{-12} . Some startup steps did not satisfy the configured outer PIMPLE criterion within three correctors. The final log must therefore be parsed before claiming complete time-step convergence or periodicity; no incomplete transient quantity is substituted by extrapolation.

Placeholder: transient residuals and convergence status per time step

Figure 13: Residual evolution for the postoperative transient simulation. Highlight any time steps that did not satisfy the selected criterion.

10.2 Velocity vectors at representative phases

Select three time points from the imposed breathing waveform, for example acceleration, peak inspiratory flow, and deceleration or flow reversal. The exact selection must be justified from the waveform rather than chosen only for visual appearance.

Placeholder: velocity vectors coloured by magnitude at three relevant time points using a common colour scale

Figure 14: Postoperative transient velocity vectors at three phases of the breathing cycle. State each time and inlet flow rate.

10.3 Breathing waveform and time-varying resistance

The applied inlet flow rate and the local resistance from Equation 13 are evaluated over one complete breathing cycle. The computational postoperative resistance is compared with the constant-diameter expectation and with a qualitative preoperative sketch.

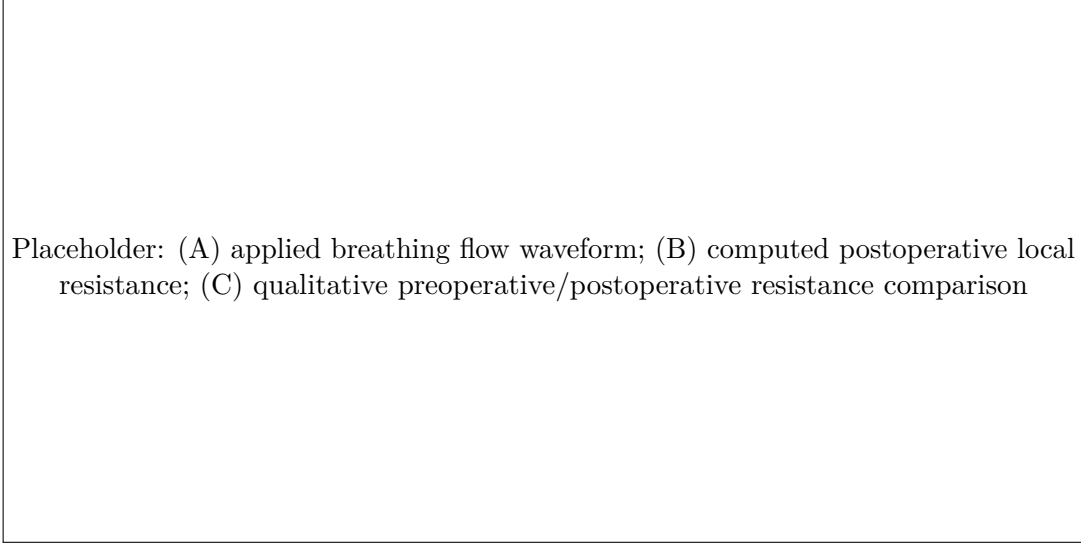


Figure 15: Breathing-cycle flow rate and local resistance. Resistance near zero flow must be handled carefully because direct division by $Q(t)$ becomes ill-conditioned.

11 Discussion

11.1 Interpretation of outlet conditions and lung-flow split

Applying the same static pressure at every distal outlet deliberately leaves the left/right flow division unconstrained, making it an output of the reconstructed geometry rather than an imposed boundary-condition ratio. The predicted 73.17% right and 26.83% left distribution is therefore a defensible measure of the relative aerodynamic resistance of the resolved postoperative branches. Differences in calibre, pathway length, branching angle, curvature, and local flow separation can all contribute to the unequal result.

This distribution should not, however, be interpreted as the child's measured physiological lung ventilation. The model ends at truncated bronchial outlets and therefore excludes the resistance of unresolved peripheral airways and the compliance of the distal lungs. A more physiologically detailed extension could apply distal resistance conditions such as

$$P_i = P_{\text{alv}} + R_i Q_i, \quad (14)$$

or coupled resistance–compliance outlet models. Such conditions would preserve an emergent flow split while representing peripheral lung mechanics, provided that patient-specific parameters were available.

11.2 Mesh sensitivity and optimal mesh

Resistance and the small right-superior branch split were more mesh-sensitive than total right-lung fraction. From 0.15 to 0.12 mm, all selected metrics changed by at most 2.02%, while cell count increased by 92% and solver execution time by 157%. The 0.12 mm case also failed residual control at 2000 iterations, whereas 0.15 mm converged in 1611. With a practical 2.5%

acceptance threshold, the 0.15 mm mesh is selected as the optimal balance between apparent spatial convergence and computational efficiency.

11.3 Constant-diameter resistance during inspiration

For fully developed laminar flow in a rigid constant-diameter tube, Poiseuille’s law gives $\Delta P = 128\mu LQ/(\pi D^4) = RQ$. Geometry and fluid properties are constant, so increasing inspiratory flow produces a proportional pressure drop while $R = \Delta P/Q$ remains constant. Patient-specific curvature, developing flow, bifurcation, separation, and inertial losses can instead make pressure drop nonlinear in flow. Transient inertia can also create phase differences between pressure and flow, so instantaneous resistance need not remain constant.

11.4 Limitations

Potential limitations include:

- static CT geometry despite the dynamic nature of tracheomalacia;
- rigid airway walls and absence of fluid–structure interaction;
- steady inlet flow instead of a respiratory waveform;
- idealised outlet pressure conditions;
- uncertainty from segmentation and clipping-point placement;
- an isotropic first-order tetrahedral mesh without dedicated prism boundary layers;
- the selected laminar-flow assumption; and
- analysis of a limited number of anatomical cases.

With additional resources, refinement should focus locally on the matched tracheal region, carina, and small right-superior branch rather than reducing global size uniformly. All cases should first meet residual and monitored-integral criteria. Additional levels around the apparent asymptotic range, consistently generated prism layers, Richardson extrapolation, and a grid-convergence index would strengthen numerical uncertainty estimates. Runtime, memory, mesh quality, and section-area variation should be tracked alongside each physical metric.

11.5 Handling transient non-convergence

[Identify unconverged time steps and discuss targeted remedies: reduce the time step, constrain the Courant number, increase inner iterations/correctors, improve mesh quality, use a better initial condition from a previous cycle, and simulate additional cycles until periodic behaviour is obtained. Report which changes would be tested first and why.]

12 Conclusions

[Summarise the principal quantitative findings, state whether the intervention improved the simulated flow characteristics, and report whether the conclusions were robust to mesh refinement.]

Reproducibility statement

The complete workflow is documented in `WORKFLOW.md`. Record the Git commit, Slicer version, Gmsh version, OpenFOAM image identifier, mesh sizes, cell counts, and solver settings used for the final results. The scripts `create_volume_mesh.sh`, `run_cfd.sh`, `run_fine_cfd.sh`, and `fetch_cfd_results.sh` automate the meshing, remote execution, reconstruction, and retrieval stages.

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